

Anatomy of a Closed-Loop Collapse: A Causal Case Study of a Compressed VLA Policy

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Abstract—Compressed manipulation policies can pass offline evaluation while failing in closed-loop execution; this offline–closed-loop dissociation is established in prior work and is not our claim. We contribute a causal anatomy of one naturally occurring case. An 8-layer distillation of Octo-Base retains 86% of parameters, passes every offline check we applied (0.996 and 1.000 teacher-ratios on the model family’s own validation metrics), and collapses in closed loop: 0/72 versus the teacher’s 40/72 on a simulated WidowX pick-and-place task. The collapse is structured, not diffuse: early task stages degrade gradually (the student still moves the object at 90% of the teacher’s rate and grasps at 55%), while transport-to-target fails categorically, at 0% in every training variant. Paired action-trace forensics isolate the signature: a negative, late-heavy z residual, roughly an order of magnitude larger than after repair, and persistent across the base distillation and both continuation branches. Four standard therapies fail under matched controls: continued training and in-domain offline data leave success at zero, even though the latter measurably improves marginal action statistics; command-level compensation recovers nothing at any offset, although the same perturbations demonstrably degrade healthy policies; clamping the symptom in the command channel preserves grasping, yet success stays at floor. A minimal-pair controlled intervention that substitutes half of the training stream with deployment-distribution teacher rollouts, with every other setting held fixed, restores parity with the teacher (18/36 vs. 17/36 held-out), eliminates the negative, late-heavy signature, and recovers a teacher-like perturbation-response profile. We claim existence, not universality. The operational implication: offline gates, including a family’s own validation metrics, are insufficient acceptance tests for compressed policies; a few dozen closed-loop trials sufficed to find what they missed.

I. INTRODUCTION

We compressed a 12-layer generalist robot policy into an 8-layer student and watched it pass every offline check we applied before deployment. The compression is mild, retaining 86% of parameters for a $1.08\times$ latency gain, and by the offline measures of its own model family the student returns teacher-level ratios: 0.996 of the teacher’s gripper-prediction accuracy, 1.000 of its translation-direction error. In closed-loop execution of a simulated WidowX pick-and-place task, the student still moves the object in 18 of 24 episodes (teacher: 20) and grasps it in 11 (teacher: 20). It transports the object to the target in **zero**, with 0/24 on the instrumented slice, 0/72 on

the full protocol, and zero again in every retrained variant we produced. The teacher scores 40/72 under an identical harness.

That gradient, from graded degradation through the early stages to a categorical cliff at the final one, is the object of this paper. The coarse phenomenon behind it, strong offline metrics coexisting with closed-loop failure, is not our discovery and we do not claim it: the offline–closed-loop gap is established for manipulation policy evaluation [8], and its theoretical substrate, compounding execution error under covariate shift, dates to imitation-learning classics [5], [6] with modern treatments [7]. What the literature does not contain, to the best of a documented search (Sec. II), is a *causal anatomy* of a naturally occurring instance: where such a failure lives, which standard remedies fail against it and why, and what suffices to cure it, all established at single-case resolution with controlled interventions.

We contribute that anatomy:

- 1) **A localized lesion.** Failure concentrates at one stage boundary (transport), and paired action traces isolate its signature, namely a negative, late-heavy z residual, roughly $10\times$ its post-repair magnitude, persistent across the base distillation and both continuation branches (Sec. IV).
- 2) **Controlled falsification of four standard therapies.** Continued training is inert. In-domain offline data is inert on success while measurably improving marginal action statistics, so whatever the deficit is, it is not those statistics. Command-level compensation recovers nothing at any offset, though the same perturbations demonstrably degrade healthy policies. The instrument works; the therapy does not; an additive-bias account is excluded. Clamping the symptom in the command channel preserves grasping and still fails, so the damage is not a single-channel defect (Sec. V).
- 3) **A sufficient therapy as a minimal pair, with completeness of recovery.** Substituting half the training stream with deployment-distribution teacher rollouts, while every other setting is held identical and verified from execution records, restores held-out success to teacher parity (18/36 vs. 17/36). Recovery is complete in a stronger sense: the negative, late-heavy trace signature is eliminated, and the repaired student exhibits a teacher-like perturbation-response profile, while the zero-rollout control stays at floor (Sec. VI).

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Our claim is existential, not universal. We do not assert that all compressed VLAs fail, nor that all such failures share this anatomy. We assert that this class of failure exists. It is mild-compression-induced, invisible to the standard offline battery, stage-localized, resistant to symptomatic and data-volume remedies, and curable by substituting successful teacher rollouts collected in the deployment environment into an otherwise unchanged training stream. We also assert that its anatomy can be established at single-case resolution with controlled interventions, the resolution existence claims require.

For evaluation infrastructure the consequence is operational. Six hundred held-out trajectories of offline evaluation returned teacher-level scores; twenty-four closed-loop rollouts separated the two policies completely, and one paired trace comparison isolated the failure’s signature. Acceptance pipelines for compressed policies need closed-loop probes, and this case shows how little they can cost, since a few dozen trials sufficed here.

II. RELATED WORK

Offline-closed-loop dissociation and evaluation. Prior manipulation-policy evaluations document that offline validation metrics can provide unreliable estimates or rankings of closed-loop performance [8]. Classical imitation-learning analyses attribute such gaps to learner-induced state-distribution shift and compounding execution error [5], [6], with a recent continuous-action treatment in [7]. The nearest prior in vocabulary, Factored Scaling Curves [9], shows that a purpose-built offline proxy, namely policy-embedding similarity, can effectively rank data-collection choices, including for generalization to unseen environment variations; ranking data-collection options and accepting a compressed policy for deployment are *different decisions*, and our case concerns the latter. We do not revisit the existence of this gap or its theoretical account; we contribute the causal anatomy of one naturally occurring, compression-induced instance.

VLA compression and distillation. A growing methods literature compresses VLA policies via architectural miniaturization [17], early-exit decoding [18], action-expert and self-derived distillation [19], [20], pruning-based recovery pipelines [15], and VLM-assisted distillation [16]. These works deliver methods, that is, smaller policies that work, whereas ours delivers a causal anatomy of why a compressed policy that should work does not. Two of them parallel our anatomy from the healthy side of the boundary. On the necessity side, [15] reports the same shape at a different depth: structured pruning collapses closed-loop success to zero, supervised fine-tuning on offline demonstrations recovers much but not all of it, and only on-policy interaction in the deployment environment completes recovery to parity. On the sufficiency side, the pipeline of [16] trains its student on teacher rollouts executed in the deployment environment, which closely matches the data regime our controlled repair identifies as sufficient, and flags success-only filtering as an untested limitation, the same confound we disclose in Sec. VII.

Failure analysis and error dynamics. SilentDrift [10] demonstrates a closely related error dynamic by attack: smooth, phase-targeted micro-perturbations implanted through training-data poisoning compound through chunked open-loop execution into task failure. That dynamic is engineered there, whereas our compression case arises naturally. Diagnostic work dissects VLA policies at other levels: representation- and pathway-level tracing attributes control to modality pathways in intact models [11]; mechanistic studies localize function inside VLA internals via activation injection and sparse autoencoders [12]; counterfactual failure synthesis generates failure-correction pairs from successful demonstrations to train recovery [13]; and gripper-state ambiguity analysis root-causes one specific failure mechanism and repairs it [14]. None jointly combines stage-level localization, action-level marker identification, and interventional testing of competing explanations for a naturally occurring, compression-induced closed-loop failure, the anatomy this paper contributes.

Novelty statement. Under a documented retrieval protocol with three channels (faceted keyword search, citation-graph traversal from seven verified seed papers, author tracking) that yielded 1,001 deduplicated candidates with tiered full-text review of top-risk items, we found no prior work satisfying the criteria above jointly.

III. THE CASE AND EVALUATION PROTOCOL

A. The compressed policy

The teacher is Octo-Base-1.5 (12 transformer layers) [1]. The student inherits 8 layers by expert surgery, which retains layers $\{0, 1, 3, 4, 6, 8, 10, 11\}$ (both boundary layers plus a uniform interior sample), and trains for 300k steps (batch 128, peak LR 3×10^{-5} reached after 2,000 warmup steps, cosine-decayed to 10^{-6}) on the Open X-Embodiment mixture [2], with a convex combination of hidden-state distillation (weight α ; cosine distance over observation, task, and readout token groups) and the diffusion action head’s native loss (weight $1-\alpha$), at $\alpha=0.5$. Training telemetry is stable through 300k steps: over the final 10k steps the distillation loss holds at ≈ 0.010 under the $1 - \cos$ definition, action loss averages 0.98 (range 0.81–1.23 across per-100-step records), and gradient norm averages 4.3 (3.6–5.3). Throughout, “compression” abbreviates this complete procedure, comprising layer removal, layer selection, the distillation objective, and the ensuing optimization; its components are never separated in this study.

The compression is deliberately mild, and we state its economics against our own interest: total parameters 202.5M $\rightarrow$ 174.1M (86% retained; the transformer stack itself compresses to 68%, diluted by shared tokenizers), mean inference latency 135.4 $\rightarrow$ 124.9 ms (1.08 $\times$). We do not advocate this configuration as an efficiency method; it is the object of study. That so mild a compression, under so unremarkable an offline profile, produces a categorical closed-loop failure is precisely what makes the case informative.

B. Offline evaluation: the acceptance gates

Offline evaluation uses the model family’s own validation utilities over held-out trajectories from the three in-distribution datasets of the training mixture (Bridge, Fractal, Kuka; 200 each): gripper-prediction accuracy and end-effector translation-direction error, with a stricter auxiliary translation metric excluded as uninformative ($< 1\%$ for both models). The student passes at teacher-ratios of 0.996 (gripping) and 1.000 (xyz angle); ratios are oriented so that 1.0 denotes teacher parity for accuracy and error metrics alike. By the gates a practitioner would apply before deployment, this distillation succeeded.

A first-pass retention sweep extends the same metrics to three exploratory datasets outside the training mixture (fridge-opening, tabletop rotation, pouring; 200 each). Across all six probes the student tracks the teacher with small, sign-mixed differences, ahead on four of the six direction-error readings, and on the battery’s best-scoring probe, fridge-opening, it matches the teacher outright: 13.3° direction error at 0.914 strict accuracy, against the teacher’s 14.2° at 0.918. Nothing in this battery distinguishes the student from a successful distillation. We use these readings as what they operationally are, namely the standard battery’s acceptance verdict, and make no offline fidelity claim beyond gate passage; per-dataset records are provided in full in the supplementary material.¹

C. Closed-loop protocol

Closed-loop evaluation runs in SimplerEnv / ManiSkill2 (SAPIEN) [3], [4] on `PutEggplantInBasketScene-v0` with a WidowX arm. Gripper actions binarize at threshold 0.75, a calibration for the Octo-1.5 model family. It was verified by a sanity run of the older Octo base and is applied identically to every policy here. The diagnostic protocol evaluates 72 configurations (seeds $\{0, 1, 2\} \times$ episode IDs 0–23); a single-seed slice carries dense instrumentation. Staged outcomes are logged by the environment’s own evaluator: object moved, grasped, sustained hold (≥ 5 consecutive steps), transport to target.² Repeatability: the teacher scores 15/24 and 14/24 on identical reruns; we attribute ± 1 to GPU floating-point nondeterminism over 120-step rollouts. All z quantities in this paper, including paired residuals, injected offsets, and clamp floors, are per-step world-frame translation commands in meters, measured after action unnormalization and identical in scale; $+z$ denotes upward motion.

For the repair phase, episode IDs partition into a selection set ($\{0-2, 9-14, 21-23\}$) and a held-out set ($\{3-8, 15-20\}$; 12 IDs $\times$ seeds $\{0, 1, 2\} = 36$ configurations). Checkpoint selection used only selection-set configurations, and was locked before any held-out evaluation; the held-out set was then evaluated exactly once per checkpoint for three bracketing checkpoints. Checkpoint identity is enforced by per-step staging directories (guarding a loader default that silently resolves to

¹<https://github.com/Xanadum/closed-loop-collapse>

²Exact predicates extracted from the environment source are provided in the metric-definitions document of the supplementary material.

TABLE I
STAGED CLOSED-LOOP OUTCOMES, INSTRUMENTED SLICE (SEED 0, EPISODES 0–23). STAGES ARE THE ENVIRONMENT EVALUATOR’S OWN PREDICATES.

Policy	Succ.	Moved	Grasp	Sust.	Target
Teacher (12L)	15/24	20	20	19	15
Student V1 (8L, 300k)	0/24	18	11	11	0
+5k OXE (Step 0)	0/24	19	16	13	0
+5k Bridge (Step 1)	0/24	20	12	10	0

the latest step), and action-normalization statistics are verified bit-identical across all compared checkpoints ($\max |\Delta| = 0$).

D. Evidence grades

Three measurement grades recur below, with different load ratings. *Counted outcomes* (36- or 24-trial success and stage counts) are robust and carry every causal claim. *Paired trace statistics* (per-episode student–teacher differences, episode composition held fixed) control initialization-level between-episode variation, improving sensitivity at small n , and carry panel-scoped, action-level diagnostic claims. *Unpaired micro-displacement differences* at the 0.002–0.006 scale are dominated by natural per-episode spread (± 0.003) at attainable n and carry *no* claims in this paper; where relevant we say so explicitly (Sec. V). This discipline is itself a finding we commend to evaluation-infrastructure practice.

IV. THE LESION: A GRADED COLLAPSE WITH A CATEGORICAL EDGE

Table I contains the paper’s founding observation, and it is a shape, not a number. Across the first three stages the student is a degraded but recognizable copy of its teacher, moving the object at 90% of the teacher’s rate, grasping at 55%, holding at 58%. At the final stage the copy ends: transport to target is 0/24, and it is 0 in *every* variant, with 0/72 on the full protocol and zero again after each retraining of Sec. V. The early stages degrade gradedly; the last fails categorically. A policy that has lost capability uniformly does not look like this; a policy whose conditional mapping is intact on some state regions and deficient on others does.

The dissociation also survives independent re-measurement: across full re-evaluations the teacher scores 40/72 (diagnostic protocol) and 36/72 (split protocol), while the student never exceeded 2/36 in any evaluation run of this project (0/72 diagnostic; 0/36 held-out; 1/36 on a selection-set anchor; 1–2/36 across perturbation sweeps). We disclose every student non-zero we observed; none approaches the teacher’s band.

Paired action-trace analysis identifies a persistent residual signature on a fixed diagnostic panel. The panel comprises four distinct hard initializations (IDs 2, 11, 14, 23), retained from a six-episode list (IDs 2, 11, 14, 15, 20, 23) fixed in the drift-instrumentation scripts before any paired-trace analysis existed: the four on which the teacher also fails, matching failure status across policies (all four lie in the selection set). The panel was held fixed for every subsequent

model comparison; V1 reaches grasp on 2 of 4. Comparisons are therefore matched by episode initialization, although the policies may occupy different states after their closed-loop trajectories diverge. Per-(episode, time-third) differencing against the teacher controls for initialization-level between-episode variation and improves sensitivity at $n = 4$; it does not make four episodes representative. We therefore scope all trace claims to this panel, while every causal claim rests on counted outcomes elsewhere.

On this panel, the mean student-teacher z residual is negative in each time-third and largest in the final third of the 120-step rollout: mean $d_z = -0.0067$ over the full rollout and -0.0103 in the final third for V1. These intervals are fixed thirds by step count, not grasp-event-aligned phases. Because V1 grasps on only 2 of 4 panel episodes, an event-aligned post-grasp analysis would rest on at most two episodes. We therefore describe the marker temporally as late-heavy and let the counted stage profile in Table I establish the link to the transport transition.

The selected marker is a negative, late-heavy z residual, roughly an order of magnitude larger than the repaired model’s residual (Sec. VI). It persists in both continuation branches on the same panel ($-0.0067 / -0.0048 / -0.0047$ over the full rollout; $-0.0103 / -0.0073 / -0.0081$ in the final third), despite continued training under two different data regimes. It is therefore a persistent pattern across these branches rather than an idiosyncrasy of one checkpoint. Its role is diagnostic, not causal: on this fixed panel it separates the three failing training variants from the repaired checkpoint. Section V then tests two simple causal accounts directly, namely a fixed additive z bias and a single-channel z defect.

Why z ? Not by magnitude, because the gripper-open marginal difference is larger. Two counted observations narrow the field first, each stated at its precise strength. Conditional grasp retention is largely preserved once a grasp is registered: all 11 of V1’s grasped episodes reach the sustained-hold criterion, as do 13 of 16 and 10 of 12 for the two continuation branches. Because the flag is latched, this establishes retention for at least 5 consecutive steps but says nothing about state-conditional misuse later in the episode. Yet no episode reaches the target in any variant. The categorical break therefore lies between sustained holding and target arrival, rather than in the mere ability to establish and briefly retain a grasp.

Step 1 (Sec. V) also changes the observed gripper and rotation marginals substantially while success remains at zero. This shows that reducing those marginal discrepancies is insufficient for recovery; it does not exclude state-conditional defects in either channel, consistent with the later evidence that the damage is not confined to one action dimension.

Among the measured action-level candidates, z is prioritized by conjunction: negative early-, late-, and full-rollout phase means, late-heavy timing, persistence across the base distillation and both continuation branches, and mechanistic relevance, since a negative teacher-relative z residual is compatible with insufficient upward lift during transport. These are criteria for selecting an intervention target, not proof of

TABLE II
PAIRED z RESIDUAL ON THE FIXED FOUR-EPIISODE DIAGNOSTIC PANEL (IDS 2, 11, 14, 23). ENTRIES ARE MEAN STUDENT-TEACHER z DIFFERENCES, COMPUTED PER EPISODE AND THEN AVERAGED ($n=4$ EPISODES PER CELL); INTERVALS ARE FIXED THIRDS OF THE 120-STEP ROLLOUT. NEGATIVE VALUES INDICATE THE STUDENT COMMANDS LESS UPWARD DISPLACEMENT THAN THE TEACHER ON THE SAME INITIALIZATION. THE BASE DISTILLATION AND BOTH CONTINUATION BRANCHES SHARE THE RESIDUAL’S SIGN IN THE EARLY-THIRD, FINAL-THIRD, AND FULL-ROLLOUT MEASURES, ALONG WITH ITS LATE-HEAVY TEMPORAL SHAPE; UNDER REPAIR THESE RESIDUALS DROP TO NEAR ZERO AND THE NEGATIVE, LATE-HEAVY PATTERN IS ELIMINATED.

Policy	Early	Mid	Late	Full
Student V1 (8L, 300k)	-0.0069	-0.0028	-0.0103	-0.0067
+5k OXE (Step 0)	-0.0064	-0.0007	-0.0073	-0.0048
+5k Bridge (Step 1)	-0.0061	+0.0001	-0.0081	-0.0047
Repaired (step 8k)	-0.0005	+0.0037	-0.0007	+0.0009

causality. The interventions of Sec. V subsequently demote even z itself: it is the most visible projection of the damage, not the damage. The closure evidence is convergent rather than independent. Under the repair of Sec. VI, the early-, late-, and full-rollout residuals drop to near zero on this same panel, while held-out success returns under the counted protocol. The two measurements move together across model conditions.

V. FALSIFIED THERAPIES: THE NECESSITY ARM

A causal claim earns its keep against alternatives. We admit a hypothesis when it has mechanistic plausibility, entanglement with the target variable in our data, and intervenability; Table III lists every admitted hypothesis with its decisive test. This section walks four therapies of increasing intervention depth, each targeting one hypothesis, each failing informatively. All verdicts below rest on counted outcomes.

A. Continued training (H1)

Step 0 resumes the student from 300k for 5,000 steps on the unchanged OXE mixture (peak LR 3×10^{-6}): the control for “it just needed more training.” Success 0/24 (moved 19, grasped 16, sustained 13, target 0); the paired signature persists ($d_z = -0.0048$ all-phase). A second, longer control appears in Sec. VI: ten thousand further steps of pure OXE under the repair-phase settings also end at 0/36.

B. In-domain offline data (H2)

Step 1 is the folk remedy: if the task domain is under-covered, add its data. A dual-stream schedule interleaves the OXE mixture with the task-domain Bridge dataset (repair batch 64 every third step, weight 0.25) for 5,000 steps from the same fork. The therapy *visibly moves marginal action statistics*. The gripper close-fraction gap shrinks from -0.161 to -0.065 , raw-open from 0.097 to 0.046, and rotation from 0.0053 to 0.0033 (paired differences on the six-episode instrumentation list of Sec. IV, collected before the repair-phase split was defined). Success stays at 0/24 with transport still zero. Improving the observed action marginals is therefore insufficient for recovery, and frames from the task-domain

TABLE III
HYPOTHESIS LEDGER. ADMISSION: PLAUSIBILITY $\times$ ENTANGLEMENT $\times$ INTERVENABILITY.

Hypothesis	Decisive test	Verdict
H1 Same-data continuation is sufficient	Step 0: continue, same data	Not supported at tested budgets (0/24; 0/36)
H2 Tested offline task-domain augmentation is sufficient	Step 1: add Bridge data	Insufficient (marginals move; success stays 0/24)
H3 Fixed additive z bias in the command channel	z -offset sweep	Excluded over tested offsets (no recovery; healthy policies degrade)
H4 Suppressing the negative z command is sufficient	Phase-gated command clamp	Falsified (grasp retained; success at floor)
H5 State-conditional policy deficiency in the deployment regime	Rollout-data minimal pair (Sec. VI)	Supported (bundle sufficient; active component unresolved)
H6 Failure solely from simulator physics	Teacher control; in-sim repair	Constrained (Sec. VII)
H7 8L capacity ceiling on this task	Same 8L reaches post-repair parity	Falsified for this task
H8 Audited harness defect explains the gap	Positive control (teacher 40/72); repeatability; audits	Excluded within audited components

dataset are not, by themselves, what is missing. Whether the broader successful-teacher-rollout bundle supplies supervision that this task-domain augmentation does not is the question Sec. VI tests.

C. Command-level compensation (H3)

The signature’s small, constant-sign, concentrated shape is what an additive-bias account predicts, and that account would deflate this paper to a calibration footnote: add $+\delta z$ at deployment and be done. It is the most deflationary competitor and must be tested first. We sweep fixed additive z -corrections over $\{0.002, 0.004, 0.006, 0.008, 0.012, 0.020\}$, applied to every commanded action throughout the rollout, 36 counted trials per offset.

No offset recovers the student: success reads 1, 0, 1, 0, 0, 0, 0 across the grid (baseline included), within the student’s observed floor range (0–2/36). The null is not an artifact of an inert instrument: the same perturbations, applied to healthy policies (Sec. VI, Table IV), degrade them monotonically; at offsets ≥ 0.008 the persistent upward bias prevents grasp formation for every policy. The instrument demonstrably alters behavior; it cannot rescue this policy at any dose. An additive bias, a correctable constant riding on an otherwise-correct policy, is excluded.

One honest limit. We attempted to quantify *how* the policy’s own commands respond to injected offsets at the trace level. At the injected scale (0.002–0.006) the per-episode natural spread of post-grasp z (± 0.003) dominates the signal, and grasp stochasticity varies the episode composition across conditions at attainable n (2–6). These measurements are included in the

supplementary material but carry no mechanism claims here. This is an instance of the evidence-grade discipline of Sec. III.

D. Command-level clamping (H4)

The offset sweep excludes only the simplest command-space account: a fixed additive z bias riding on an otherwise correct policy. It does not determine whether the negative z command itself is an operative part of the failure, because the policy’s subsequent closed-loop commands can counteract a constant correction. We therefore replace the correction with a constraint that cannot be counteracted in the command space: a phase-gated hook clamps the commanded z -delta to a floor, engaging once the policy first issues a gripper-close command, a command-level phase marker that precedes, and does not require, the evaluator’s physical grasp predicate. The gate works: grasping is preserved (16–17/36) and the commanded net z -delta after gate engagement turns non-negative by construction ($-0.0066 \rightarrow +0.0012 / +0.0031$). Success does not return: 2/36 at the unclamped base, 0/36 and 2/36 under the two clamp settings, the student’s floor. Trajectories shift laterally (mean commanded x displacement after gate engagement roughly doubles; small- n trace observation, reported qualitatively). With the headline symptom forbidden in the command channel, the policy still cannot transport. The damage is not a single-channel z defect; the drift is the most visible projection of a deficient mapping, not the deficiency itself.

E. The intervention-depth gradient

Read as a ladder (Fig. 1): command-level correction fails at every dose; the hard command constraint is circumvented; retraining on the unchanged mixture is inert at 5k and at 10k steps; retraining with offline task-domain data is inert on success while active on marginals. All four rungs fail. The arrow points upstream, from output-level symptom correction into the training-time supervision that formed the conditional mapping; Sec. VI intervenes there with a matched substitution of the successful-teacher-rollout bundle.

VI. SUFFICIENT THERAPY UNDER MINIMAL-PAIR CONTROL

A. Deployment-distribution data

We collect 324 successful teacher rollouts in the deployment environment via environment-variable-gated recording hooks, under configuration-level isolation: recording uses the selection-set episode IDs ($\{0-2, 9-14, 21-23\}$) under recording seeds $\{100-179\}$, disjoint from the evaluation seeds $\{0, 1, 2\}$. The seed governs only the policy’s action-sampling stream, while the initial scene is a deterministic function of the episode ID alone; hence no recorded (seed, ID) pair coincides with any of the 72 diagnostic configurations, and the held-out episode IDs ($\{3-8, 15-20\}$) are never recorded under any seed. Episodes are packaged as an RLDS dataset (38,880 frames; the count is independently confirmed by the trainer’s own configuration record) and verified end to end: whole-stack image fingerprinting confirms 324/324 unique episodes with

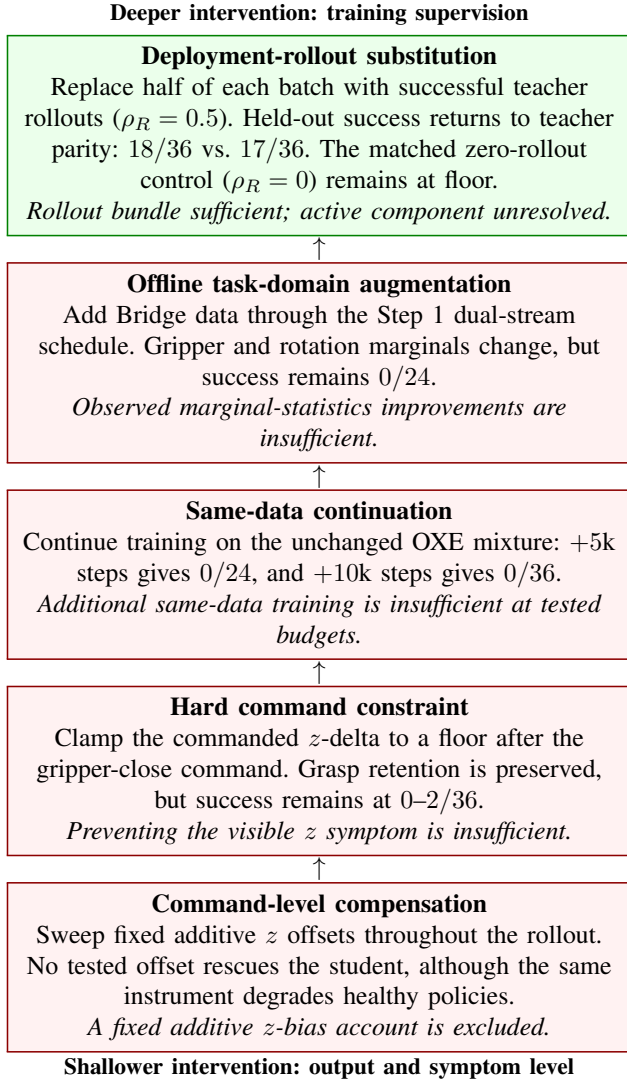


Fig. 1. **The therapy ladder.** Interventions are ordered from shallow output-level corrections at the bottom to upstream changes in training supervision at the top. The four lower rungs fail under counted closed-loop trials, ruling out progressively richer simple explanations. Only the top rung restores teacher parity (18/36 vs. 17/36 held-out); its matched zero-rollout control stays at floor, attributing the recovery to the substituted data rather than the additional optimization. This establishes the rollout-data bundle as sufficient for recovery, not any individual component within that bundle.

zero round-trip error; action-normalization statistics are bit-identical to the student’s ($\max |\Delta| = 0$).

B. The minimal pair, from execution records

Two runs of the *same* training script resume from the V1 300k checkpoint. Their execution records (experiment-tracking config and command-line metadata, included in the supplementary material) agree on every setting, namely seed (42), learning rate (constant 10^{-6} , warmup 200, no decay), optimizer state reset, loss (the original distillation objective at $\alpha=0.5$; no task-specific or z -specific terms added), total batch (128), steps (10,000), checkpoint cadence (every 500), fork point, host, and GPU. They differ in exactly one argument:

$\rho_R = 0.5$ versus $\rho_R = 0$, i.e., whether half of each batch is drawn from the rollout dataset (64+64) or none is (128+0).

Two properties of this design carry weight. First, it is *substitution, not addition*: the repair run receives no extra gradient budget, steps, or sample presentations, because half its OXE stream is replaced by deployment-distribution data at identical total volume. A larger update or sample-presentation budget is unavailable as an explanation; the recovery must arise from the content of the substituted stream. Second, the repair stream is small and reused: 38,880 frames revisited ≈ 13 times by step 8,000. We disclose this; the held-out result below argues against memorization of the recorded episodes. One argument is not one scientific factor: the substituted stream bundles several co-varying properties (Sec. VII).

C. Outcome: parity, reported against selection

Checkpoint selection used the selection set only, where step 8,000 was the in-band maximum (24/36), and was locked before any held-out evaluation. Held-out was then evaluated once per checkpoint for the three bracketing checkpoints: 20/36 (step 6k), **18/36** (step 8k), 21/36 (step 10k). We report the pre-selected checkpoint’s 18/36 against the teacher’s 17/36 under the identical harness, which we read as *parity, not improvement* because the difference is within run-to-run repeatability (Sec. III), and we decline the higher neighboring number that post-hoc selection would have claimed. The selection-set margin over the teacher (24/36 vs. 19/36, Table V) is consistent with specialization to the recorded scenes, because the selection IDs deterministically reproduce the very layouts on which the rollouts were recorded, which is one more reason the held-out comparison, on never-recorded layouts, carries the report. Grasping does not regress: held-out grasp 34/36, against the unrepaired student’s 17/36, so the cure is not purchased by forgetting. The zero-control run stays at floor at every checkpoint (single exception: 1/36 at step 8k): the recovery is attributable to the substituted rollout data rather than to the additional optimization, and which property of that data does the work is the open question of Sec. VII.

D. Recovery on every measured axis

Parity of a single scalar can hide an unhealed policy. Two further measurements say the recovery is complete in kind, not only in rate, on every axis we measured.

The negative, late-heavy signature is eliminated. On the same paired 4-episode base as Sec. IV, the repaired student’s d_z is +0.0009 all-phase and -0.00065 late, an order-of-magnitude reduction from V1’s -0.0067 / -0.0103 , and the early-third residual falls comparably ($-0.0069 \rightarrow -0.0005$). The mid-third, where the failing signature was never consistently expressed, does not follow the failing pattern (Table II). The repaired model grasps 4/4 on a base where V1 managed 2/4.

The perturbation-response profile follows the teacher’s. Under the same z -offset instrument of Sec. V, the repaired student’s counted dose-response curve shows a dose-dependent degradation similar to the teacher’s, with 22, 22, 14, 3, 0

TABLE IV

DOSE RESPONSE UNDER COMMAND-LEVEL PERTURBATION. COUNTED SUCCESS (OUT OF 36) UNDER FIXED ADDITIVE z OFFSETS APPLIED THROUGHOUT THE ROLLOUT, EVALUATED ON THE CHECKPOINT-SELECTION CONFIGURATIONS (HELD-OUT CONFIGURATIONS WERE NEVER USED FOR PERTURBATION SWEEPS). THE TEACHER AND THE REPAIRED STUDENT SHOW SIMILAR DOSE-DEPENDENT DEGRADATION; THE ZERO-ROLLOUT CONTROL AND V1 REMAIN AT FLOOR AT EVERY DOSE. THE SAME INTERVENTION THAT DEMONSTRABLY DEGRADES HEALTHY POLICIES DOES NOT RESCUE EITHER FAILING ONE (SECS. V, VI). ZERO-OFFSET ENTRIES ARE SEPARATE PERTURBATION-SWEEP EVALUATIONS, NOT PART OF THE CHECKPOINT-SELECTION COMPARISON OF TABLE V.

Policy	Fixed additive z offset						
	0	.002	.004	.006	.008	.012	.020
Teacher (12L)	19	16	8	3	0	0	0
Repaired (8k)	22	22	14	3	0	0	0
Zero-control (8k)	0	2	2	1	0	0	0
Student V1	1	0	1	0	0	0	0

TABLE V

PARITY UNDER A PRE-COMMITTED SELECTION RULE. STEP 8,000 WAS SELECTED ON THE SELECTION SET ONLY (36 CONFIGURATIONS), WHERE IT WAS THE IN-BAND MAXIMUM, AND LOCKED BEFORE ANY HELD-OUT EVALUATION; FOR THE CHECKPOINT-SELECTION COMPARISON, THE HELD-OUT SET (36) WAS THEN EVALUATED ONCE PER CHECKPOINT. THE PRE-SELECTED CHECKPOINT SCORES 18/36 HELD-OUT VERSUS THE TEACHER’S 17/36 UNDER AN IDENTICAL HARNESS; THE HIGHER NEIGHBORING 21/36 AT STEP 10K IS NOT USED FOR REPORTING. HELD-OUT GRASP (34/36 VS. THE UNREPAIRED STUDENT’S 17/36) SHOWS THE RECOVERY DOES NOT TRADE AWAY GRASP PERFORMANCE. [†]PRE-SELECTED.

Policy	Selection success	Held-out success	Held-out grasp
Student V1 (8L, 300k)	1/36	0/36	17/36
Teacher (12L)	19/36	17/36	30/36
Repaired, step 6k	20/36	20/36	34/36
Repaired, step 8k[†]	24/36	18/36	34/36
Repaired, step 10k	20/36	21/36	33/36

against 19, 16, 8, 3, 0 over offsets $\{0, .002, .004, .006, .008\}$, while the zero-control stays at floor throughout (0, 2, 2, 1, 0). The repaired model’s higher absolute counts at low doses mirror its zero-offset advantage on these selection configurations (22 vs. 19), consistent with the selection-set gap in Table V; the claim is the shared degradation shape, not superiority. Parity extends from a scalar to a behavioral profile: the healed policy responds to perturbation like its teacher, not like its unhealed twin (Table IV).

E. The causal statement

The interventions support a proximate cause stated at the system level, while leaving its training-time formation and representational origin unresolved:

The proximate cause of the student’s closed-loop failure is a deficiency of its learned, state-conditional action mapping, expressed most strongly in the late transport regime and compounded by closed-loop

execution into categorical failure at the transport stage.

Each element is anchored as follows. *State-conditional, localized*: the graded-to-categorical stage profile (a policy that moves the object at 90% and transports at 0% has not lost capability uniformly) and the economy of the cure itself (38,880 frames at unchanged budget). *Conditional, not marginal*: Step 1 shifts the measured marginals without restoring success. *Not a single channel*: the most visible projection is the small, negative, late-heavy z residual of Sec. IV, compounded over the remaining rollout [5], [6], [7], yet the clamp shows the deficiency is not confined to it. Nor is it explained by a fixed additive z bias, additional same-data training at tested budgets, an audited harness defect, or an 8-layer capacity ceiling. Two controls bound compression’s role: the teacher, trained on the same offline corpus, never develops the deficiency, and the same 8-layer network recovers to parity once re-supervised. The compression-and-distillation procedure is therefore an enabling condition, not a capacity verdict; the representational route by which it enables is the layer we do not reach. The state-conditional character and the closed-loop amplification are system-level inferences from these interventions, not direct measurements of the internal mapping.

The minimal pair supplies the positive causal evidence at this functional level: under the matched protocol, substituting the successful-teacher-rollout bundle restores held-out success to teacher parity (18/36 vs. 17/36), eliminates the negative, late-heavy trace signature, and recovers a teacher-like perturbation-response profile, while the zero-rollout control remains at floor. The failure is a repairable defect in the learned policy, not an immutable architectural limitation. What the design does not identify is the active component within the bundle: the partial constraints of Sec. VII disfavor purely visual-domain, task-content, and reduced-OXE readings, leaving deployment-state coverage, success-only filtering, teacher-consistent labels, or their combination unresolved. We read the pattern as most consistent with missing supervision on the states the deployed student actually reaches, the classical covariate-shift account [5], [6], and label this an interpretation, not a measurement.

The logic survives at this grain: the remedy names none of the cause, and the cause, functionally stated, predicts the remedy class. A controlled supplement establishes that a deficiency exists and where it is expressed; it does not, by itself, name the missing nutrient.

VII. THREATS TO VALIDITY AND LIMITATIONS

Simulator-artifact hypothesis. Could the z -drift be a contact-physics artifact rather than a policy pathology? Two observations constrain this. The teacher runs under identical contact physics and does not exhibit the paired signature; and the cure operates entirely within the same simulator by changing training data alone, whereas a policy-independent physics artifact would not be expected to disappear under data-only retraining. A residual caveat remains: the student’s earlier actions could steer it into contact configurations where

artifacts trigger preferentially. We therefore scope the claim to the policy under this simulator’s dynamics; hardware transfer is future work, for which a physical WidowX platform is in place.

Capacity. Post-repair parity of the same 8-layer network falsifies a capacity ceiling for this task. Capacity may still shape which pathologies compression induces; that question is representational.

Representational origin. What the pruned layers lost such that the mapping fails on exactly these states is the layer we do not reach; the anatomy is deliberately behavioral and systemic.

The bundle confound. The repair stream bundles four properties, namely deployment-distribution states, success-only filtering, teacher-consistent labels, and the simulator’s visual domain, all of which fully co-vary in our data; ρ_R additionally covaries with reduced OXE share. The minimal pair establishes the bundle as the difference-maker, not any component. Partial constraints: stage-specificity with relatively preserved early stages disfavors a purely visual-domain reading, and Step 1’s failure on genuine task-domain data disfavors a task-content reading; the pure-OXE condition was sampled at two budgets (5k, 10k), both at floor, disfavoring the reduced-OXE reading. Component-level arms (failure-rollout admixture, student-state relabeling) are future work. The same success-only confound is flagged untested in [16]; we disclose it.

Measurement-scale limits. Micro-displacement intervention analysis at the 0.002–0.006 scale is not decision-grade at attainable n in this setting (Sec. III); counted outcomes and paired statistics carry all claims here. We commend the distinction to evaluation practice.

Scope. One model family, one task, one simulator; claims are existential accordingly. A cheap second testbed was sought and not found: within this harness the teacher itself performs at 1/12 and 0/12 on the two candidate Bridge tasks (and 0/8 on a third), so no in-family task could host the dissociation, itself an infrastructure observation: teacher competence bounds the supply of cheap testbeds.

VIII. CONCLUSION

There exists a class of compression-induced closed-loop failure that is invisible to a model family’s own offline battery, graded in its early stages and categorical at its final one, resistant to symptomatic correction at every dose and to data-volume remedies at matched budgets, and curable to teacher parity, with the negative, late-heavy signature eliminated and a teacher-like perturbation-response profile recovered, by substituting successful teacher rollouts collected in the deployment environment into an otherwise unchanged training stream. We established this at single-case resolution with a four-rung necessity ladder and a minimal-pair sufficiency arm verified from execution records. For scalable robot-learning infrastructure the implication is operational: offline gates are insufficient as standalone acceptance tests for compressed policies, and the decisive evidence was cheap, since a few dozen closed-loop trials exposed the categorical collapse, one paired trace comparison isolated its signature, and together they found what

every offline gate we applied missed. The anatomy, protocols, and controls are offered as a template for the next case.

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